Access dataframes using sql

Objectives

1. Create temporary views on dataframes
2. Access the view from SQL cell
3. Access the view from Python cell

```
%run "../includes/configuration"
```

```
race_results_df = spark.read.parquet(f"{presentation_folder_path}/race_results")
```

Local Temporary View

```
race_results_df.createOrReplaceTempView("v_race_results")
```

```
%sql
select *
from v_race_results
where race_year = 2020
```

Table

	race_year	race_name	race_date	circuit_location	driver_name	driver_nationality
1	2020	Bahrain Grand Prix	2020-11-29T14:10:00.000+00:...	Sakhir	Lewis Hamilton	British
2	2020	Bahrain Grand Prix	2020-11-29T14:10:00.000+00:...	Sakhir	Max Verstappen	Dutch
3	2020	Bahrain Grand Prix	2020-11-29T14:10:00.000+00:...	Sakhir	Alexander Albon	Thai
4	2020	Bahrain Grand Prix	2020-11-29T14:10:00.000+00:...	Sakhir	Lando Norris	British
5	2020	Bahrain Grand Prix	2020-11-29T14:10:00.000+00:...	Sakhir	Carlos Sainz	Spanish
6	2020	Bahrain Grand Prix	2020-11-29T14:10:00.000+00:...	Sakhir	Pierre Gasly	French
7	2020	Bahrain Grand Prix	2020-11-29T14:10:00.000+00:...	Sakhir	Daniel Ricciardo	Australian
8	2020	Bahrain Grand Prix	2020-11-29T14:10:00.000+00:...	Sakhir	Valtteri Bottas	Finnish
9	2020	Bahrain Grand Prix	2020-11-29T14:10:00.000+00:...	Sakhir	Esteban Ocon	French
10	2020	Bahrain Grand Prix	2020-11-29T14:10:00.000+00:...	Sakhir	Charles Leclerc	Monegasque
11	2020	Bahrain Grand Prix	2020-11-29T14:10:00.000+00:...	Sakhir	Daniil Kvyat	Russian
12	2020	Bahrain Grand Prix	2020-11-29T14:10:00.000+00:...	Sakhir	George Russell	British
13	2020	Bahrain Grand Prix	2020-11-29T14:10:00.000+00:...	Sakhir	Sebastian Vettel	German
14	2020	Bahrain Grand Prix	2020-11-29T14:10:00.000+00:...	Sakhir	Nicholas Latifi	Canadian

340 rows

```
p_race_year = 2019
```

```
race_results_2019_df = spark.sql(f"select * from v_race_results where race_year = {p_race_year}")
```

```
race_results_2019_df = spark.sql("select * from v_race_results where race_year = 2019")
```

```
display(race_results_2019_df)
```

Table

	race_year	race_name	race_date	circuit_location	driver_name	driver_nationality
1	2019	Hungarian Grand Prix	2019-08-04T13:10:00.000+00:...	Budapest	Lewis Hamilton	British
2	2019	Hungarian Grand Prix	2019-08-04T13:10:00.000+00:...	Budapest	Max Verstappen	Dutch
3	2019	Hungarian Grand Prix	2019-08-04T13:10:00.000+00:...	Budapest	Sebastian Vettel	German
4	2019	Hungarian Grand Prix	2019-08-04T13:10:00.000+00:...	Budapest	Charles Leclerc	Monegasque
5	2019	Hungarian Grand Prix	2019-08-04T13:10:00.000+00:...	Budapest	Carlos Sainz	Spanish
6	2019	Hungarian Grand Prix	2019-08-04T13:10:00.000+00:...	Budapest	Pierre Gasly	French
7	2019	Hungarian Grand Prix	2019-08-04T13:10:00.000+00:...	Budapest	Kimi Räikkönen	Finnish
8	2019	Hungarian Grand Prix	2019-08-04T13:10:00.000+00:...	Budapest	Valtteri Bottas	Finnish
9	2019	Hungarian Grand Prix	2019-08-04T13:10:00.000+00:...	Budapest	Lando Norris	British
10	2019	Hungarian Grand Prix	2019-08-04T13:10:00.000+00:...	Budapest	Alexander Albon	Thai
11	2019	Hungarian Grand Prix	2019-08-04T13:10:00.000+00:...	Budapest	Sergio Pérez	Mexican
12	2019	Hungarian Grand Prix	2019-08-04T13:10:00.000+00:...	Budapest	Nico Hülkenberg	German
13	2019	Hungarian Grand Prix	2019-08-04T13:10:00.000+00:...	Budapest	Kevin Magnussen	Danish
14	2019	Hungarian Grand Prix	2019-08-04T13:10:00.000+00:...	Budapest	Daniel Ricciardo	Australian

420 rows

Global Temporary View

Objectives

1. Create Global views on dataframes
2. Access the view from SQL cell
3. Access the view from Python cell
4. Access the view from another Notebook

```
race_results_df.createOrReplaceGlobalTempView("gv_race_results")
```

```
%sql
show tables
```

Table

	database	tableName	isTemporary
1		v_race_results	true

1 row

```
%sql
show tables in global_temp
```

Table

	database	tableName	isTemporary
1	global_temp	gv_race_results	true
2		v_race_results	true

2 rows

```
%sql
select * from global_temp.gv_race_results
```

Table

	race_year	race_name	race_date	circuit_location	driver_name	driver_nationality
1	2009	Spanish Grand Prix	2009-05-10T12:00:00.000+00:...	Montmeló	Jenson Button	British
2	2009	Spanish Grand Prix	2009-05-10T12:00:00.000+00:...	Montmeló	Rubens Barrichello	Brazilian
3	2009	Spanish Grand Prix	2009-05-10T12:00:00.000+00:...	Montmeló	Mark Webber	Australian
4	2009	Spanish Grand Prix	2009-05-10T12:00:00.000+00:...	Montmeló	Sebastian Vettel	German
5	2009	Spanish Grand Prix	2009-05-10T12:00:00.000+00:...	Montmeló	Fernando Alonso	Spanish
6	2009	Spanish Grand Prix	2009-05-10T12:00:00.000+00:...	Montmeló	Felipe Massa	Brazilian
7	2009	Spanish Grand Prix	2009-05-10T12:00:00.000+00:...	Montmeló	Nick Heidfeld	German
8	2009	Spanish Grand Prix	2009-05-10T12:00:00.000+00:...	Montmeló	Nico Rosberg	German
9	2009	Spanish Grand Prix	2009-05-10T12:00:00.000+00:...	Montmeló	Lewis Hamilton	British
10	2009	Spanish Grand Prix	2009-05-10T12:00:00.000+00:...	Montmeló	Timo Glock	German
11	2009	Spanish Grand Prix	2009-05-10T12:00:00.000+00:...	Montmeló	Robert Kubica	Polish
12	2009	Spanish Grand Prix	2009-05-10T12:00:00.000+00:...	Montmeló	Nelson Piquet Jr.	Brazilian
13	2009	Spanish Grand Prix	2009-05-10T12:00:00.000+00:...	Montmeló	Kazuki Nakajima	Japanese
14	2009	Spanish Grand Prix	2009-05-10T12:00:00.000+00:...	Montmeló	Giancarlo Fisichella	Italian

10,000+ rows | Truncated data due to row limit

```
spark.sql("select * from global_temp.gv_race_results").show()
```

```
-------+-----------+------+--------+--------------------+
|     2009|Spanish Grand Prix|2009-05-10 12:00:00|       Montmeló|       Jenson Button|           British|      Brawn|   1|
17|1:37:19.202|  10.0|       1|2024-07-21 19:41:...|
|     2009|Spanish Grand Prix|2009-05-10 12:00:00|       Montmeló|  Rubens Barrichello|         Brazilian|      Brawn|   3|
28|   +13.056|   8.0|       2|2024-07-21 19:41:...|
|     2009|Spanish Grand Prix|2009-05-10 12:00:00|       Montmeló|         Mark Webber|        Australian|   Red Bull|   5|
18|   +13.924|   6.0|       3|2024-07-21 19:41:...|
|     2009|Spanish Grand Prix|2009-05-10 12:00:00|       Montmeló|    Sebastian Vettel|            German|   Red Bull|   2|
41|   +18.941|   5.0|       4|2024-07-21 19:41:...|
```

```
29|    +52.312|   2.0|       7|2024-07-21 19:41:...|
|     2009|Spanish Grand Prix|2009-05-10 12:00:00|       Montmeló|        Nico Rosberg|           German|   Williams|   9|
24|  +1:05.211|   1.0|       8|2024-07-21 19:41:...|
|     2009|Spanish Grand Prix|2009-05-10 12:00:00|       Montmeló|      Lewis Hamilton|          British|    McLaren|  14|
29|         \N|   0.0|       9|2024-07-21 19:41:...|
|     2009|Spanish Grand Prix|2009-05-10 12:00:00|       Montmeló|          Timo Glock|           German|     Toyota|   6|
44|         \N|   0.0|      10|2024-07-21 19:41:...|
```